CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: https://en.wikipedia.org/wiki/Global_warming
Title: Climate change - Wikipedia
Word Count: 28,299 words | **Date:** 2026-10-19 10:55:53

VERDICT: FAKE

Calibrated Confidence: 93.9%
Model Disagreement (std): 0.101
Gating Trigger: None

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	67.0%	FAKE	67.0%
XGBoost	69.7%	FAKE	69.7%
LightGBM	81.5%	FAKE	81.5%
Logistic Regression	92.3%	FAKE	92.3%
CAGE-Fake Stack	93.9%	FAKE	93.9%

Consensus Status: Strong Consensus
Inter-Model Standard Deviation: 0.1112 (Threshold Delta = 0.20)
Vote Distribution: 5 FAKE vs 0 REAL
Diagnostic: Models strongly agree on the verdict with minor variance.

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	100.0/100	Excellent
Source Attribution	0.0/100	Deceptive / Alert
Epistemic Grounding	100.0/100	Excellent
Punctuation Discipline	100.0/100	Excellent
Readability Structure	100.0/100	Excellent
Lexical Diversity	70.0/100	Adequate

Overall Credibility Index: 72.0/100
Credibility Tier: B (Standard News)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:
1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **FAKE** with **93.9% confidence**.
- **Model Consensus**: Strong Consensus across base learners (5/5 models voted FAKE, average P(FAKE)=80.9%).
- **Driving Deceptive Markers**: Heavy reliance on linguistic patterns associated with misleading content (p_lgbm, p_rf, p_lr).
- **Credibility Score**: Article credibility index is 72.0/100 (B (Standard News)).

Counterfactual Simulation: Neutralizing sensational language and punctuation reduced FAKE confidence by 3.3% (from 93.9% to 90.5%), though underlying semantic patterns still lean deceptive.
Sensitivity Delta: P(FAKE) changed from 93.8% to 90.5% (-3.3%) when neutralizing sensational tokens.

4. Local Linguistic Signal Attributions

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact
p_lgbm	+10.457	feature_6	-5.319

p_rf	+3.819	feature_5	-3.374
p_lr	+2.594	feature_10	-1.806
feature_7	+2.171	feature_11	-1.348

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8639	0.8635	0.1548	100.0%	0.8639
XGBoost	0.8770	0.8769	0.0380	100.0%	0.8770
LightGBM	0.8689	0.8688	0.0418	100.0%	0.8689
Logistic Regression	0.8967	0.8967	0.0381	100.0%	0.8967
Uncalibrated Stack	0.8984	0.8983	0.0330	100.0%	0.8984
CAGE-Fake (Calibrated + Gated)	**0.9000**	**0.8999**	**0.0362**	**84.4%**	**0.9476**

6. Research Calibration & Disagreement Visualizations







